COURSE SYLLABUS

CSC11006 – INTRODUCTION TO CLOUD COMPUTING

1. GENERAL INFORMATION

Course name:	Introduction to Cloud Computing
Course name (in Vietnamese):	Nhập môn Điện toán đám mây
Course ID:	CSC11006
Knowledge block:	Specialised
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	Unlimited
Prerequisite:	Introduction of Information Technology, Introduction of Programming, Computer Networking
Prior course:	Advanced Computer Networking.
Instructors:	

2. COURSE DESCRIPTION

This course provides a comprehensive introduction to cloud computing concepts, technologies, and applications. It equips students with the knowledge and skills to understand, utilize, and evaluate cloud-based solutions for various computing needs.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Understand and explain fundamental cloud computing concepts, service models, deployment models, and the benefits and challenges of cloud adoption.	1.3.4, 2.4.3, 2.4.5
G2	Explain and apply cloud architecture principles, core cloud services (compute, storage, databases, networking), and connectivity mechanisms to design scalable and reliable cloud solutions.	1.1.1, 1.1.2, 1.2.2, 1.3.2, 1.3.4, 1.4.13,

G3	Analyze and evaluate security, privacy, identity, and compliance challenges in cloud environments, and propose appropriate mitigation strategies following industry best practices.	4.1.2, 4.3.4
G4	Apply and evaluate monitoring, elasticity, high availability, caching, automation, and infrastructure-as-code practices to build resilient and efficient cloud systems.	1.3.4, 1.4.13
G5	Analyze, design, and evaluate cloud-native architectures, including decoupled systems, serverless computing, and microservices, to develop scalable and maintainable applications.	1.3.6, 1.4.13, 4.3.4, 5.1.1, 5.1.2, 5.1.3, 6.2.2, 6.2.3
G6	Analyze and apply big data concepts, cloud-based analytics platforms, and data pipelines to support large-scale data processing and decision-making	5.1.1, 5.1.3, 5.2.1, 5.2.2, 5.3.1, 6.1.1, 6.2.2, 6.2.3

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Explain the characteristics of cloud computing, the concept of service models and deployment models of cloud computing Explain the key differences between on-premise, cloud-based, and hybrid approaches. Identify the benefits and challenges of cloud adoption	T
G1.2	Analyze and compare different cloud service models (IaaS, PaaS, SaaS) and deployment models (public, private, hybrid).	T,U
G1.3	Identify the security considerations for each cloud service model: Recognize the shared responsibility model in cloud security and understand the specific security concerns and best practices associated with each service model	T
G1.4	Identify the benefits and limitations of each cloud deployment model: Compare the agility, scalability, security, and cost considerations for public, private, and hybrid clouds.	T

G2.1	Explain the core components of cloud architecture (hypervisors, VMs, containers, storage, networks, orchestration) and their interactions.	T,U
G2.2	Analyze cloud-native architectures, including decoupled systems, serverless computing, and microservices, to develop scalable and maintainable applications.	T,U
G2.3	Design scalable and reliable cloud solutions using core services (Compute, Storage, Network).	T, U
G2.4	Evaluate existing cloud infrastructures against the AWS Well-Architected Framework for optimization.	T, U
G3.1	Analyze identity and access management mechanisms and apply the shared responsibility model in cloud environments.	T,U
G3.2	Evaluate security risks related to data protection, access control, and network exposure, and propose appropriate mitigation strategies.	T,U
G3.3	Analyze and evaluate privacy, compliance, and governance challenges in cloud computing and recommend solutions aligned with regulations and best practices.	T,U
G4.1	Apply monitoring, elasticity, and high-availability mechanisms to build resilient cloud systems.	T,U
G4.2	Analyze performance and cost trade-offs when applying caching and scaling strategies in cloud environments.	T,U
G4.3	Apply and evaluate infrastructure automation and Infrastructure as Code practices to improve system reliability and operational efficiency.	T,U
G5.1	Analyze the principles of decoupled, event-driven, serverless, and microservices architectures.	T,U
G5.2	Design and implement cloud-native applications using messaging services, serverless computing, and container technologies.	T,U
G5.3	Assess the impact of microservices and serverless architectures on application scalability.	T,U
G6.1	Analyze big data concepts and cloud-based analytics platforms for large-scale data processing and decision making.	T,U

G6.2	Apply cloud analytics services to design basic data pipelines and analytics workflows.	T,U
G6.3	Apply and analyze cloud services through hands-on labs and projects to solve real-world computing problems.	T, U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduction to Cloud Computing: • Definition and evolution of cloud computing • Benefits and challenges of cloud adoption • Cloud service models (IaaS, PaaS, SaaS) • Deployment models (public, private, hybrid) • Major cloud service providers (AWS, Azure, GCP, etc.) • Case studies of cloud computing in various industries	G1.1, G1.2	Activities: • Discussion	Quiz 1
2	Cloud Architecture and Virtualization • Cloud infrastructure and components • Virtualization technologies (hypervisors, VMs, containers) • Resource management and orchestration • High availability and scalability in the cloud • Cloud monitoring and cost management	G2.1, G2.2, G2.3, G2.4, G3.1	Activities: • Teaching • Demo • Discussion	Quiz 2

	• AWS Well-Architected Framework • Best practices for building solutions on AWS • AWS Global Infrastructure			
3	Securing Access • Security principles • Authentication and securing access • Authorization users • IAM policy	G3.1, G3.2, G3.3	Activities: • Teaching • Case study • Demo • Discussion	Quiz 3 Guide Lab_03: Exploring AWS Identity and Access Management
4	Adding a Storage Layer with Amazon S3 • Defining Amazon S3 • Using Amazon S3 • Moving Data to and from Amazon S3 • Storing content with Amazon S3 • Designing with Amazon S3 • Applying the AWS Well-Architected Framework principles to storage	G2.1, G3.2, G4.1	Activities: • Teaching • Case study • Demo • Discussion	Quiz 4 Challenge Lab_04: Creating a Static website for the cafe.
5	Adding a Compute Layer Using Amazon EC2 • Add compute with Amazon EC2 • Choosing an AMI to launch an EC2 instance • Selecting an EC2 instance style • Adding storage to an Amazon EC2 instance	G2.1, G4.1, G4.2	Activities: • Teaching • Case study • Demo • Discussion	Quiz 5 Guided Lab_05: Introducing Amazon Elastic File System (Amazon EFS) Challenge Lab_05: Creating a Dynamic

	• Other EC2 configuration considerations • Amazon EC2 pricing options • Applying the AWS Well-Architected Framework principles to compute			Website for the Cafe'
6	Adding a Database Layer • Database layer consideration • Amazon RDS • Amazon RDS proxy connection management • Amazon DynamoDB • Purpose-built databases • Migrating data into AWS databases • Applying the AWS Well-Architected Framework principles to database layer	G2.1, G4.1, G4.2	Activities: • Teaching • Case study • Demo • Discussion	Quiz 6 Guided Lab_06: Creating an Amazon RDS Database Challenged Lab_06: Migrating a Database to Amazon RDS
7	Creating a Network Environment • Introduction to Amazon VPC • Securing network resources • Connecting to managed AWS services • Monitoring your network • Applying AWS Well-Architected Framework principles to a network	G2.1, G3.2	Activities: • Teaching • Case study • Demo • Discussion	Quiz 7 Guided Lab_07: Creating a Virtual Private Cloud Challenged Lab_07: Creating a VPC Networking Environment for the Cafe
8	Connecting Networks	G2.1, G3.2	Activities: • Teaching • Case study	Quiz 8

	● Scaling your VPC network with AWS Transit Gateway ● Connecting VPCs in AWWS with VPC peering ● Connecting to your remote network with AWS Site-to-Site VPN ● Connecting to your remote network with AWS Direct Connect ● Applying AWS Well-Architected Framework principles to network connectivity		● Demo ● Discussion	Guided Lab_08: Creating a VPC Peering Connection.
9	Securing User, Application, and Data Access ● Managing permissions ● Federating users ● Managing access to multiple accounts ● Encrypting data at rest ● AWS security services for securing user, application, and data access	G3.1, G3.2, G3.3	Activities: ● Teaching ● Case study ● Demo ● Discussion	Quiz 9 Guided Lab_09: Securing Applications by using Amazon Cognito Guided lab_09.2: Encrypting Data at Rest by Using AWS Encryption Options
10	Implementing Monitoring, Elasticity, and High Availability ● Monitoring your resources ● Scaling your compute resources ● Scaling your databases ● Using load balancers to create highly available environments	G2.4, G4.1, G4.2	Activities: ● Teaching ● Case study ● Demo ● Discussion	Quiz 10 Guided Lab_10: Creating a Highly Available Environment Challenge Lab_10: Creating a Scalable and Highly Available

	● Using Amazon Route 53 to create highly available environments ● Applying AWS Well-Architected framework principles to highly available systems			Environment for the Café
11	Automate Your Architecture ● Reasons to Automate ● Using infrastructure as code ● Customizing with CloudFormation ● Using AWS Quick Starts ● Customizing With Amazon Q Developer ● Applying AWS Well-Architected framework principles to automation	G4.3	Activities: ● Teaching ● Case study ● Demo ● Discussion	Quiz 11 Guided Lab_11: Automating Infrastructure with AWS CloudFormation Challenge lab_11: Automating Infrastructure Deployment
12	Caching Content ● Overview of caching ● Caching using CloudFront ● Caching using ElastiCache ● Applying the AWS Well-Architected Framework principles to caching	G4.1, G4.2	Activities: ● Teaching ● Case study ● Demo ● Discussion	Quiz 12
13	Building Decoupled Architectures	G5.1, G5.2	Activities: ● Teaching ● Case study	Quiz 13

	● Decoupling your architecture ● Decoupling applications with Amazon SQS ● Decoupling applications with Amazon SNS ● Decoupling a hybrid application with Amazon MQ		● Demo ● Discussion	Guided lab_13: Building Decoupled Applications by Using Amazon SQS
14	Building Serverless Architectures and Microservices ● Thinking serverless ● Architecting serverless microservices ● Building serverless architectures with AWS Lambda ● Building microservice applications with AWS container services ● Orchestrating microservices with AWS Step Functions ● Extending serverless architectures with Amazon API Gateway ● Applying AWS Well-Architected framework principles to microservices and serverless architectures ●	G2.4, G5.1, G5.2, G5.3	Activities: ● Teaching ● Case study ● Demo ● Discussion	Quiz 14 Guided Lab_14: Implementing a Serverless Architecture on AWS Guided lab_14.2 Breaking a Monolithic Node.js Application into Microservices Challenge lab_14: Implementing a Serverless Architecture for the Café
15	Big Data and Analytics in the Cloud	G6.1, G6.2	Activities: ● Teaching ● Case study	Quiz 15

	• Introduction to big data concepts and challenges • Big data analytics platforms and services in the cloud • Cloud-based data warehousing and data lakes • Data visualization and machine learning in the cloud • Case studies of big data analytics in the cloud		• Demo • Discussion	
16	Planning for Disaster on Cloud • Disaster planning strategies • AWS disaster recovery planning • Disaster recovery patterns • Applying AWS Well-Architected framework principles to disaster planning	G3.3, G4.1	Activities: • Teaching • Case study • Demo • Discussion	Quiz 16 Guided Lab_16: Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3 File Gateway

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
EX1	Quiz	Quiz 1 - Quiz 16	G1.1, G1.2, G1.3, G1.4, G2.1	5% (Extra credit)

GLab	Guided Lab		G2.4, G3.1, G3.2, G3.3, G4.1, G4.2, G4.3, G5.1, G5.2, G5.3, G6.1	10%
CLab	Challenge Lab		G2.4, G3.1, G3.2, G3.3, G4.1, G4.2, G4.3, G5.1, G5.2, G5.3, G6.1	10%
PR	Project			30%
P2	Project 01	Big Data Pipeline with Predictive Analytics / Cloud-Based Log Analytics and Business Intelligence System	G2.1, G2.2, G2.3, G3.1, G6.2, G6.3	15%
P3	Project 02	Deploy and Manage a Scalable Cloud-Native E-Commerce Platform	G2.1, G2.2, G2.3, G2.4, G3.1, G5.2, G5.3	15%
Exam	Midterm Exam	Quiz		15%
	Final Exam	Quiz		35%

7. RESOURCES

Textbooks

[1]. Jill West. (2020). *CompTIA Cloud+ Guide to Cloud Computing (MindTap Course List)* (1st ed.), Cengage Learning

[2]. "Cloud Computing: A Practical Approach" by Anthony T. Veloso

[3]. "Cloud Computing: Principles and Paradigms" by Rajkumar Buyya, James Broberg, and Andrzej Goscinski

[4].

Tools

[5]. VMWare

[6]. Kahoot

[7]. Public Clouds: AWS, Azure, Google Cloud

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.
- Students prepare lessons, preview documents according to regulations
- Students need to actively interact in online discussion environments
- All online accounts must be registered by student email, using the student-ID and full name, the real avatar in online workspace.
- The number of assignments may vary depending on the classroom situation